

1 Iterative Methods for Solving Linear Systems

1.1 List of Solvers

1. Conjugate Gradient (CG)
2. BiConjugate Gradient (BiCG)
3. BiCGSTAB
4. Stone's SIP (Standalone Solver)
5. Preconditioned CG (PCG)
6. Preconditioned BiCG
7. Preconditioned BiCGSTAB

1.2 Nomenclature and Variables

A Coefficient matrix of the linear system.

Q Right-hand side source vector.

$\phi^{(k)}$ Approximate solution vector at iteration k .

$\rho^{(k)}$ Residual vector at iteration k .

$p^{(k)}$ Search direction vector at iteration k .

$z^{(k)}$ Auxiliary vector at iteration k .

$\alpha^{(k)}, \beta^{(k)}, \xi^{(k)}$ Scalar coefficients.

1.3 1. Conjugate Gradient (CG) — Without Preconditioner

Applicable only to Symmetric Positive Definite (SPD) systems.

Initialization

$$\begin{aligned}k &= 0 \\ \phi^{(0)} &= \phi_{\text{in}} \\ \rho^{(0)} &= Q - A\phi_{\text{in}} \\ p^{(0)} &= 0 \\ s^{(0)} &= 10^{30}\end{aligned}$$

Iteration Advance counter:

$$k = k + 1$$

Compute:

$$s^{(k)} = \rho^{(k-1)} \cdot \rho^{(k-1)}$$

$$\beta^{(k)} = \frac{s^{(k)}}{s^{(k-1)}}$$

$$p^{(k)} = \rho^{(k-1)} + \beta^{(k)} p^{(k-1)}$$

$$\alpha^{(k)} = \frac{s^{(k)}}{p^{(k)} \cdot Ap^{(k)}}$$

$$\phi^{(k)} = \phi^{(k-1)} + \alpha^{(k)} p^{(k)}$$

$$\rho^{(k)} = \rho^{(k-1)} - \alpha^{(k)} Ap^{(k)}$$

Convergence Check Stop if

$$\|\rho^{(k)}\| < \varepsilon.$$

Else continue iterating.

1.4 2. BiConjugate Gradient (BiCG) — Without Preconditioner

Applicable to nonsymmetric systems.

Initialization

$$k = 0$$

$$\phi^{(0)} = \phi_{\text{in}}$$

$$\rho^{(0)} = Q - A\phi_{\text{in}}$$

$$\bar{\rho}^{(0)} = Q - A^T \phi_{\text{in}}$$

$$p^{(0)} = 0$$

$$\bar{p}^{(0)} = 0$$

$$s^{(0)} = 10^{30}$$

Iteration Advance counter:

$$k = k + 1$$

Compute:

$$\begin{aligned}
s^{(k)} &= \rho^{(k-1)} \cdot \bar{\rho}^{(k-1)} \\
\beta^{(k)} &= \frac{s^{(k)}}{s^{(k-1)}} \\
p^{(k)} &= \rho^{(k-1)} + \beta^{(k)} p^{(k-1)} \\
\bar{p}^{(k)} &= \bar{\rho}^{(k-1)} + \beta^{(k)} \bar{p}^{(k-1)} \\
\alpha^{(k)} &= \frac{s^{(k)}}{\bar{p}_k \cdot A p^{(k)}} \\
\phi^{(k)} &= \phi^{(k-1)} + \alpha^{(k)} p^{(k)} \\
\rho^{(k)} &= \rho^{(k-1)} - \alpha^{(k)} A p^{(k)} \\
\bar{\rho}^{(k)} &= \bar{\rho}^{(k-1)} - \alpha_k A^T \bar{p}^{(k)}
\end{aligned}$$

Convergence Check Stop if

$$\|\rho_k\| < \varepsilon.$$

1.5 3. BiCGSTAB (GPBi-CG) — Without Preconditioner

Initialization

$$\begin{aligned}
k &= 0 \\
\phi_0 &= \phi_{\text{in}} \\
\rho_0 &= Q - A\phi_{\text{in}} \\
\bar{\rho}_0 &= \rho_0 \\
p_0 &= 0 \\
\alpha_0 &= 10^{30} \\
\xi_0 &= 10^{30} \\
\beta_1 &= 0
\end{aligned}$$

Iteration Advance counter:

$$k = k + 1$$

Compute:

$$\begin{aligned}
p_k &= \rho_{k-1} + \beta_k (p_{k-1} - \xi_{k-1} A p_{k-1}) \\
\alpha_k &= \frac{\rho_{k-1} \cdot \bar{\rho}_0}{(A p^{(k)}) \cdot \bar{\rho}_0} \\
z_k &= \rho_{k-1} - \alpha_k A p_k \\
\xi_k &= \frac{(A z_k)^T z_k}{(A z_k)^T (A z_k)} \\
\phi_k &= \phi_{k-1} + \alpha_k p_k + \xi_k z_k \\
\rho_k &= z_k - \xi_k A z_k \\
\beta_{k+1} &= \frac{\xi_k}{\alpha_k} \frac{\rho_k^T \hat{\rho}_0}{\rho_{k-1}^T \hat{\rho}_0}
\end{aligned}$$

Convergence Check Stop if

$$\|\rho_k\| < \varepsilon.$$

1.6 4. Stone's SIP

1.6.1 Nomenclature

Grid and Indices

i, j : Grid coordinates

N_j : Number of grid points in j -direction

$$l = (i - 1)N_j + j$$

Neighbor indexing:

North $\rightarrow l + 1$

South $\rightarrow l - 1$

East $\rightarrow l + N_j$

West $\rightarrow l - N_j$

Coefficient Matrix For a five-point stencil:

$$A_P, A_E, A_W, A_N, A_S$$

represent the central, east, west, north and south coefficients.

ILU Matrices Lower triangular matrix:

$$L_W, L_S, L_P$$

Upper triangular matrix:

$$U_E, U_N$$

with unit diagonal.

Stone Parameter

$$0 < \alpha \leq 1$$

where $\alpha = 0$ gives ILU(0).

1.6.2 Initialization

$$n = 0$$

$$\phi^{(0)} = \phi_{\text{in}}$$

Compute SIP coefficients:

$$\begin{aligned}
L_W^l &= \frac{A_W^l}{1 + \alpha U_N^{l-N_j}} \\
L_S^l &= \frac{A_S^l}{1 + \alpha U_E^{l-1}} \\
L_P^l &= A_P^l + \alpha \left(L_W^l U_N^{l-N_j} + L_S^l U_E^{l-1} \right) \\
&\quad - L_W^l U_E^{l-N_j} - L_S^l U_N^{l-1} \\
U_N^l &= \frac{A_N^l - \alpha L_W^l U_N^{l-N_j}}{L_P^l} \\
U_E^l &= \frac{A_E^l - \alpha L_S^l U_E^{l-1}}{L_P^l}
\end{aligned}$$

1.6.3 Iteration

Advance:

$$n = n + 1$$

Residual

$$\begin{aligned}
\rho_l &= Q_l - A_P^l \phi_l - A_N^l \phi_{l+1} - A_S^l \phi_{l-1} \\
&\quad - A_E^l \phi_{l+N_j} - A_W^l \phi_{l-N_j}
\end{aligned}$$

Forward Substitution Solve:

$$LR = \rho$$

$$R_l = \frac{\rho_l - L_S^l R_{l-1} - L_W^l R_{l-N_j}}{L_P^l}$$

Backward Substitution Solve:

$$U\delta = R$$

$$\delta_l = R_l - U_N^l \delta_{l+1} - U_E^l \delta_{l+N_j}$$

Update

$$\phi_l^{\text{new}} = \phi_l + \delta_l$$

Convergence Check Compute residual norm:

$$\sum_l |\rho_l|$$

Stop if

$$\|\rho\| < \varepsilon.$$

Otherwise set

$$\phi \leftarrow \phi^{\text{new}}$$